FLIR

skywatch™



USER GUIDE

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W0004 Operation Manual_rev R

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WELCOME TO THE TEAM

From all of us at FLIR Integrated Systems, congratulations on your decision to become a part of the SkyWatch crime deterrence and surveillance system network. We are dedicated to ensuring protection and peace of mind for people wherever SkyWatch products are being used. This guide is designed to help you obtain the highest level of satisfaction from your SkyWatch tower. It provides information regarding safety precautions, operations and troubleshooting, plus additional information in case you need to contact us. It is our goal to make sure that you are completely satisfied with your SkyWatch product. Once again, we want you to know how pleased we are that you are a part of our efforts to improve quality of life by providing the very best products in crime deterrence and surveillance. We are glad you are part of our team!

COMPANY MISSION

Our goal at FLIR Integrated Systems is to change the way that security and surveillance are done.

MARKET VALUE PROPOSITION

The core value proposition of FLIR Integrated Systems is Customer Intimacy. This is an approach characterized by strong customer relationships, high customization and strong collaboration with the customer.

NEED SUPPORT?

As part of the FLIR Integrated Systems team, you have access to a skilled support staff available to assist you in any situation.

CUSTOMER SUPPORT CONTACT NUMBERS

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1.0 SAFETY PRECAUTIONS

1.1 MECHANICAL

Before operating your SkyWatch Tower, read this User Guide and become familiar with it and with your equipment. Safe and efficient operation can be achieved only if the equipment is properly operated and maintained. Many accidents can be prevented by to following fundamental rules and regulations.

Fuels, electrical equipment, batteries, exhaust fumes and moving parts present potential hazards that could result in serious personal injury. Take care in following these recommended procedures. Be sure to review the hazards found in each of the accessory equipment owners' manuals in addition to this section. These manuals are included on your Technical Reference CD.

1.1.1 FUEL AND GASOLINE FUMES

Refer to the generator owner's manual for hazards associated with fuel usage.

Do NOT store open or unsealed fuel containers on the tower while the on-board generator is in operation or while the Tower is being transported.

Do NOT smoke or use an open flame around the on-board generator or fuel tank.

1.1.2 EXHAUST FUMES

These gases are deadly. The on-board generator engine exhaust fumes contain carbon monoxide, a gas that can be potentially fatal.

Do NOT operate SkyWatch in any type of enclosed area, while utilizing the on-board generator. To operate SkyWatch indoors, power is supplied to the tower via power cords. See Section 6.5 Shore Power, for more details.

1.1.3 MOVING PARTS

Improper operation can cause severe personal injury or death.

Keep hands away from any moving parts or pinch points.

Before descending, check to make sure nothing is in the way of the Officer's Compartment descent path.

1.2 ELECTRICAL

Electrical shock can cause severe personal injury or death.

- Turn off the main power switch before attempting to service any of the Tower's electrical equipment. Use rubber insulated mats, placed on dry wood platforms over floors that are metal or concrete, when around electrical equipment.
- Do NOT wear damp clothing (particularly wet shoes) or allow exposed skin to be damp when handling electrical equipment.
- Use extreme caution when working on electrical equipment. High voltages can cause injury or death.
- Follow state and local electrical codes. Have all installations or electrical connections performed by qualified, licensed electricians. Tag all open switches to avoid accidental closure and shock.

1.3 GENERAL

- Benzene and lead, found in some gasoline, have been identified by certain state and/or federal agencies as causing cancer or reproductive toxicity. When checking, draining or adding fuel to your SkyWatch Tower, take care not to ingest, to breathe the fumes or to allow direct skin contact.
- Used engine oils have been identified by some state and/or federal agencies as causing cancer or reproductive toxicity. When checking or changing engine oil, take care not to ingest or to allow direct skin contact with used oil.
- Do NOT work on this equipment when mentally and/or physically fatigued or after consuming alcohol, drugs or medications that may impair safe operation of the tower.

- Your SkyWatch Tower has been designed for security and surveillance purposes only. Do NOT use the Tower for anything that would compromise the safety of personnel or of the tower.
- Depending on model, your SkyWatch Tower may or may not include Anti-Ballistic capabilities. If your SkyWatch Tower does not include such capabilities, then no claims or assertions of Anti-Ballistic capability are implied. Proper assessment of the threat level of any situation or area must be made prior to deployment. Failure to comply with this directive may result in injury to personnel and damage to the Tower for which FLIR Integrated Systems assumes no responsibility.
- FLIR Integrated Systems is not responsible for injury or damage resulting from non-standard equipment installations or modifications added to the Tower by anyone other than FLIR Integrated Systems factory authorized personnel.

CAUTION! Do NOT deploy this Tower under power lines, electrical or communication wires. The ascent path of the Officer's Compartment may come in contact with these wires and other obstructions. Make sure the Tower is not set up within fifty (50) feet of any electrical lines.

CAUTION! Do NOT deploy this Tower on sloping or uneven terrain.

CAUTION! Do NOT deploy this Tower in water. This Tower is designed and intended to be deployed only on a firm surface. If wet or rainy conditions persist, a proper assessment must be made regarding the necessity to relocate the tower.

WARNING: Do not attempt to deploy, use or operate any part of this Tower until this User Guide has been read thoroughly!

2.0 GLOSSARY



Breakaway Control Device: This safety feature is designed to engage the electrical brakes on the SkyWatch Tower in the event the Tower becomes separated from the towing vehicle while in transit. The key, which activates the breakaway brakes, is located on the side of the tongue. When the key is removed the brakes are engaged and will stay engaged until the key is replaced. Failure to replace this key will drain the on-board battery.



GFI Outlets: These outlets are designed to automatically open the circuit should they become overloaded or grounded in water.



Outriggers: These are used to give the Tower its stability and are necessary for safe deployment. During deployment, the outriggers are to be locked into the open position for maximum safety and stability.



Safety Chains: This safety measure is used during the transport of the Tower. These chains keep the Tower connected to the tow vehicle in case the Tower becomes uncoupled during transport.



Split-ball Coupler: This is designed to secure the Tower to the tow vehicle



Outrigger Lock Pins: These pins lock the outriggers into place during deployment and also during transport.



Stabilizer Jacks: These are required to secure the Tower and help provide maximum safety and stability when properly deployed. These jacks are at the end of each of the outriggers.



Stabilizer Jack Adjuster Pins: These are utilized to allow the height of the stabilizer jack to be adjusted quickly.



Tail Light Hookup: This links the electrical system of the tow vehicle to the SkyWatch Tower during transport.



Tongue Mounted Jack: This is designed to lift the Tower off the tow vehicle and is also used for aiding in leveling the Tower for deployment.

3.0 TOW VEHICLE REQUIREMENTS

SkyWatch Tower Model	Truck Size	Hitch Size	Ball Size	Gross Weight	Tongue Weight
Sentinel	¾ Ton	Class IV	2 5/16"	9,000 lbs	800 - 1,200 lbs
Standard Frontier	¾ Ton	Class IV	2 5/16"	9,500 lbs	1,000 - 1,200 lbs
Level III A	1 Ton	Class IV or V	2 5/16"	10,000 lbs	1,000 - 1,200 lbs
Level III	1 Ton	Class IV or V	2 5/16"	11,000 lbs	1,100 - 1,200 lbs
Level IV	5 Ton	Class V	2 5/16"	15,000 lbs	1,400 - 1,700 lbs

This section describes the minimum requirements needed to safely tow the SkyWatch Tower that you have purchased from FLIR Integrated Systems. Along with these requirements, you must check with your state and local authorities to verify that the vehicle and tow equipment meet their requirements.



Tow Vehicle: ¾ ton or heavier vehicle (pick-up truck or SUV) for standard models only



Must use one (1) ton or larger for Anti-Ballistic models



Class IV or heavier hitch: With a 2 5/16" tow ball hitch

NOTE: A Class IV hitch is a frame mounted receiver style hitch. This is not the same as a bumper hitch.



7 Pin Receiver Plug: For turn signals, running lights, brake lights and electric brakes



Electric Brake Controller: All SkyWatch Towers are equipped with electric brakes. The electric brake controllers are generally sold as an aftermarket item for the tow vehicle.



Outside Rear View Mirrors: Must be wide enough to see the rear corners of the SkyWatch Tower (Approximately 22 feet from the bumper of the tow vehicle)



Safety Chain Hook Receivers: Generally incorporated in the hitch



Figure 1 2012 SkyWatch Sentinel

4.0 UNCOUPLING AND COUPLING

4.1 UNCOUPLING SKYWATCH FROM TOW VEHICLE

Your SkyWatch Tower is equipped with an easily operated split-ball coupler. Sections 4.1 and 4.2 cover the function and operation of the hitch and tongue-mounted jack. Refer to these sections for directions on coupling and uncoupling the tow vehicle. It is recommended that you become familiar with this section before attempting to couple or uncouple your SkyWatch Tower. Failure to do so may result in personal injury, death or damage to the unit.

Uncoupling Instructions

Step 1: Chock one wheel on each side, using the wheel chocks provided.

Step 2: Disconnect the 7 pin connector tail light hookup, remove safety chains and remove the breakaway brake cable from tow vehicle.

Step 3: Turn the tongue jack handle clockwise until the foot touches the ground.

Step 4: Release the split-ball coupler.

- Remove the coupler's safety pin.
- Grip the retaining sleeve with two hands and pull back (away from the tow vehicle) until the spring-loaded coupler releases.

Step 5: Return to the tongue jack and continue to rotate the handle (clockwise) until the coupler is released from the hitch.

Step 6: Drive tow vehicle away slowly from the Sky Watch Tower.

Step 7: Proceed on to stabilizing the Tower. (See Section 5.0)

4.2 COUPLING SKYWATCH TO TOW VEHICLE

Coupling Instructions

Step 1: Make sure the Officer's Compartment door is closed and locked.

Step 2: Raise the stabilizer jack feet to their limit.

Step 3: Raise the stabilizer jack extensions, by pulling the stabilizer jack adjusting pins out and lifting the extension, then releasing the pins.

Step 4: Using the tongue jack, adjust the height of the tongue for connection to the tow vehicle. Proceed to back the tow vehicle under the split-ball coupler.

Step 5: When the hitch is centered below the split-ball coupler, stop the vehicle.

Step 6: Rotate the jack handle counter clockwise until the coupler is in contact with the $2^{5/16}$" tow ball.

Step 7: Using your foot, press the split-ball coupler's right side until the locking collar deploys around the coupler.

Step 8: Continue to rotate the jack handle counter clockwise until the jack comes to a stop.

Step 9: Make Tower as level as possible, especially on two-axle models.

Step 10: Place the safety pin in the coupler.

Step 11: Attach the safety chains, using the cross chain method, to the tow vehicle.

Step 12: Attach the 7 pin connector tail light hookup.

Step 13: Check the tail light operation.

Step 14: Remove the wheel chocks.

WARNING: Test trailer brakes. Make sure safety chains and running lights are connected to the towing vehicle before attempting to tow the Tower.

CAUTION! Check light hook-up for full function prior to transport. When storing any equipment on the Tower, keep it secured and centered in front on the axle for best towing results.

WARNING: Observe all posted speed limits while towing.

WARNING: Do not put any additional equipment inside the Officer's Compartment.

5.0 SETUP

WARNING: Do not attempt to elevate your SkyWatch Tower until all outriggers and adjustable stabilizers are properly deployed. Failure to comply may result in personal injury or death. Make sure the Tower tires are properly chocked before attempting to elevate the Tower.

CAUTION! Make sure the Tower is deployed on firm and level surfaces. Avoid soft or uneven surfaces. For maximum stability, the Tower should be fully elevated or fully lowered during deployment. When the Tower is deployed, the area around the Tower should be roped or coned off for added safety.

5.1 STABILIZING PROCEDURES

Step 1: Make sure that one wheel is chocked on each side of the Tower.

Step 2: Deploy the outriggers to the outermost limits.

- Pull the locking pin up.
- Rotate the outrigger to its outermost limits.
- Make sure the locking pin is in place, pin will lock itself into place.
- Pull stabilizer jack adjuster pin. The extension will drop until at maximum extension or contact with ground. Release extension pin.
- Rotate the stabilizer jack handle clockwise. Continue until the extension pin engages the extension.
- Repeat these steps on the remaining outriggers.

Step 3: Ensure the outriggers are correctly deployed and in firm contact with the ground.

Step 4: Using the installed leveling bubbles, level the Tower using the stabilizer jacks to place the bubbles in between the lines.

Step 5: The SkyWatch Tower is now stabilized.

5.2 PREPARING FOR OPERATIONS

Step 1: If operating the Tower on generator power, make sure that there is appropriate fuel in the fuel tank and start the generator.

Step 2: Unlock the Electrical Compartment, open the left door.

Step 3: On the left side of the Compartment, locate the rotary power selector switch. If using Shore Power, push in on the selector and then twist it, clock-wise, into the SHORE POWER position. If using the on-board generator, push in on the selector and then twist it, counter clock-wise into the ONAN POWER position.

- At this time, the SkyWatch Tower is ready for operation with the Officer's Compartment resting on the transportation gate. This is an acceptable location to operate the Tower from. Continue the following steps for operation from ground level.

Step 4: On the left side of the Electrical Compartment, locate the orange control pad (See Section 6.0). Press and hold the UP button until the Officer's Compartment is approximately five feet off the ground.

Step 5: Walk to the Officer's Compartment. Just past the fender, stop and lift the transportation gate until it is resting on the rubber bumpers of the crossbar.

CAUTION! The transportation gate is heavy. Get assistance to prevent injuring yourself.

Step 6: From the Electrical Compartment, on the orange control pad, press and hold the DOWN button until the Officer's Compartment is completely resting on the ground. At this time, the Tower is ready for operation from ground level.

6.0 OPERATION

Your SkyWatch Tower is designed to elevate one officer from ground level to an eye level of approximately 24 feet, depending on the model. This section covers the operation of the Tower. It is recommended that the operator be completely familiar with the Tower for safe operation.

CAUTION! Certain feature components on your SkyWatch Tower have their own, specific operation instructions which need to be read thoroughly by all personnel. These components include the generator, air conditioner, heater, thermal imaging camera, DVR and fire extinguisher.

6.1 PRE-OPERATION CHECKS

Before operating the Tower, check the ascent (up) path of the Officer's Compartment and perform a brief inspection to make certain the Tower is ready for operation. Give the entire system a visual inspection for loose parts, severed cables, oil leaks, hydraulic leaks or fuel leaks. Repair any problems prior to operating the Tower.

6.2 ELECTRICAL COMPARTMENT COMPONENTS



Up/Down Momentary Switch: The operator can control the Tower's ascent and descent from the technical compartment.

To raise: Press and hold the UP button until the Officer's Compartment has reached the maximum elevation. The Compartment will stop moving when it has reached maximum elevation, the pump will continue to run until the button is released.

To lower: Press and hold the DOWN button until the Officer's Compartment has reached ground level. The Compartment will stop moving when it has reached ground level. Release the button.

NOTE: The Officer's Compartment door must be closed for the Tower to operate.



Power Selector Switch: This switch allows the operator to select the on-board generator (ONAN POWER), or shore power, which is an external 110 Volt power source.



110 Volt GFI Outlet: Provides 110 Volt power. Generator or shore power must be selected for this outlet to operate.

6.3 OFFICER'S COMPARTMENT COMPONENTS



A/C Controls: The thermostat controls are located overhead with easy to read temperature settings.

NOTE: The A/C should never be in operation during ascent



Switches 5 & 6: These switches activate the roof-mounted spotlights. The number of the switch corresponds to the number on the roof end panels for each light. To activate a light, flip the toggle switch up. To turn off the light, flip the toggle switch down.

UP/DOWN



Joystick / Safety Button: The safety button MUST be depressed while the Tower is in operation. The joystick controls the ascent and descent of the Officer's Compartment. To elevate the Compartment, with safety button compressed, lift and hold the joystick until the Compartment stops moving. When no movement is felt, release the joystick. To descend, with the safety button depressed, push the joystick down and hold until the Compartment stops moving, then release the joystick.



Generator Start/Stop Switch: To start the on-board generator, press and hold the rocker switch in the up position until the generator is running, then release. To shut down the generator, press and hold the rocker switch in the down position until the generator has stopped running, then release.



12 Volt DC Outlet: This outlet provides 12 Volts DC at 10 amps to power hand held spotlights, cellular phones, radios or any accessories that requires 12 Volt power.



Fuel Gauge: This gauge is to keep the operator informed of the fuel level in the fuel tank.



Interior Light Switch: This switch is used to provide illumination inside the Officer's Compartment.



Floodlights (1-4): Each SkyWatch Tower is equipped with four floodlights, one located above each window. The lights come on in pairs. Each of these switches controls one pair of lights. The numbers on the switch labels correspond to the numbers found above each window, inside the Compartment.



110 VAC Outlet (GFI): This outlet is located on the right side of the control panel. This outlet is equipped with Ground Fault Interruption. This outlet is designed to automatically open the circuit should it be overloaded or should a direct short occur.



Weather Station: This sensor measures various meteorological conditions – the most important being the wind speed.

6.4 ELECTRICAL OPTIONS

This section refers to the two (2) 110 Volt AC power sources available. These sources are an on-board generator and shore power (external power). It is essential that the operator be familiar with the electrical considerations covered in this section, for safe and efficient operation of the 110 Volt accessories. Read and understand this section thoroughly before operating.



On-Board Generator: To operate the SkyWatch Tower using the on-board generator, rotate the power selector switch to ONAN POWER.

CAUTION! Do NOT operate the SkyWatch Tower in an enclosed building while running the on-board generator!

6.5 SHORE POWER

To operate the Tower using shore power, rotate the power selector switch (located in the Electrical Compartment) to SHORE POWER. Using a heavy duty (12 gauge or heavier) grounded extension cord, plug the Tower into a dedicated (single use) 110 Volt circuit rated at 20 amps or better.

Shore power is required when operating the SkyWatch Tower indoors.

NOTE: Do NOT attempt to plug too many 110 Volt accessories into the on-board outlets. The simultaneous operation of too many accessories could overload the circuit and cause the generator circuit breaker to trip.

6.6 LIFT FUNCTIONS

CAUTION! The Officer's Compartment door must be closed and the safety button depressed for the SkyWatch Tower to elevate from either lift location.

6.6.1 FROM THE TECHNICAL COMPARTMENT

Step 1: Verify that the power selector switch is set for power source that you are using (either ONAN POWER or SHORE POWER).

Step 2: To elevate the Officer's Compartment, depress the UP button on the up/down momentary switch box. Continue to depress button until the Officer's Compartment is fully elevated. Release the button.

Step 3: To lower the Officer's Compartment, push the DOWN button on the up/down momentary switch box. Continue to depress button until Officer's Compartment has stopped movement. Release the button.

6.6.2 FROM THE OFFICER'S COMPARTMENT

Step 1: Verify that the power selector switch is set for power source that you are using (either ONAN POWER or SHORE POWER).

Step 2: Open a window prior to closing the door. The Compartment has a tight seal, and opening a window will allow the air to escape and the door to close and latch easier.

Step 3: Close and secure the Officer's Compartment door.

Step 4: Attach the door safety cable; this cable prevents the door from fully opening while the Compartment is in the raised position.

Step 5: To elevate the Tower, on the control panel, depress the safety button and lift the joystick until the Compartment has reached its maximum elevation. Release the joystick. The Tower is now ready for operation.

Step 6: To lower the Tower, on the control panel, depress the safety button and push the joystick until the Compartment has ceased movement. Release the joystick.

7.0 TROUBLESHOOTING

Problem	Probable Cause	Solution
Fails to ascend	Officer's Compartment door is not closed	Close and secure the door
Moaning and vibrations (usually during ascent)	Unit has not been properly lubricated Cylinder seal has dried out	Refer to Section 9.0: Scheduled Maintenance for SkyWatch Apply penetrating oil where rod moves out of cylinder
Electrical systems not operating	Circuit has been overloaded Generator is not running Shore power cord not plugged into an outlet	Reset generator circuit breaker. Reset circuit breakers in technical compartment Restart generator Plug cord into an outlet. Verify outlet circuit not overloaded
Difficultly in towing	Tower not set in tow mode Excess weight in Officer's Compartment Tires not properly inflated	Verify unit is in tow mode. Refer to Section 4.2: Coupling SkyWatch to Tow Vehicle Verify that extra items are not stowed in Officer's Compartment Verify tires are properly inflated, refer to Section 9.0: Maintenance – Tires for proper inflation information

8.0 MAINTENANCE

As a matter of your daily routine, you should visually inspect your SkyWatch Tower for any obvious signs of mechanical damage, leaking fluids, electrical system problems or other maintenance issues. While your SkyWatch Tower has been designed to require minimal daily and weekly routine maintenance, careful attention to your Tower's condition can help prevent minor irritations from becoming major headaches.

8.1 LUBRICATION SYSTEM

Your SkyWatch Tower is equipped with grease fittings at each pivoting joint. These grease fittings are located on the top and bottom of each outrigger (8) and at the pivot joints of both upper arms and lower arms (total of 14). These joints are lubricated prior to shipment to you. Proceed with regular lubrication as indicated in the maintenance schedule.

8.2 BATTERY CARE

The start battery is used to crank the generator. This battery is charged by the generator and by the installed solar panel. It is important to keep your solar power clean and free from dust to allow for optimal start battery charging capabilities. There is a status light in the Electrical Compartment that will show if the solar panel is charging the start battery. If this light is not lit, check the condition of your solar panel.

8.3 HYDRAULIC SYSTEM

The fill cap is located on the top of the pump reservoir. Only add fluid if the Tower does not elevate to full height. Contact your FLIR Integrated Systems Customer Support Center to determine if this is the cause.

CAUTION! Use only Dextron II transmission fluid. Reservoir should be filled to within ¼" of the reservoir top. Only measure the reservoir when the Officer's Compartment is on the ground, not on the transport gate. Measuring and filling the reservoir when the Officer's Compartment is on the transport gate will overfill the reservoir.

8.4 FLOW CONTROL VALVE

This valve is factory set so that the descent of a fully elevated cab to the ground should take approximately 90 seconds. An end user may need to adjust this valve due to any of the following conditions.

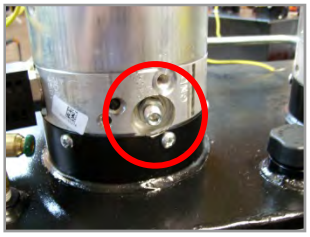


- Temperature
- Altitude
- Amount of hydraulic fluid in system
- Excessive humidity
- Normal mechanical wear on components

We recommend twisting the adjustment knob no more than one half turn at a time. Make the adjustment, then test the descent rate and make further adjustments, as necessary. We highly recommend that faster descent rates not be used.

8.5 PUMP FLOW RATE ADJUSTMENT (2)

CAUTION! This fitting should never be adjusted by the end user!



These adjustments (one located on each pump) are set at the factory to provide optimal integration with the rest of the components in the hydraulic system. Incorrect calibration of this component can cause:

- Hydraulic motor failure due to excessive back pressure from the pump
- Cylinder failure due to excessive hydraulic fluid pressure on the cylinder seals

8.6 GENERATOR

Refer to the generator operator's manual for details about maintenance on your on-board generator. Your SkyWatch Tower is equipped with an oil drain hose and valve for quick and convenient oil changes.

CAUTION! Make sure that the oil drain valve is securely closed before refilling with oil.

8.7 GENERAL

Establish and adhere to a definite schedule of maintenance and service. If the Tower is subjected to extreme operating conditions, consult a FLIR customer support specialist, to determine the appropriate maintenance schedule.

Keep an accurate log of all maintenance and service, for warranty support.

Perform all maintenance at the operational hours as indicated. Use this schedule to determine the maintenance required.

On the following pages you will find a detailed preventative maintenance schedule. To keep your SkyWatch Tower operating at peak efficiency, please follow this maintenance schedule. Remember to always keep a detailed log of any maintenance issues that you may have with your Tower and also detail the steps taken to correct those issues. Finally, keep records of all performed scheduled maintenance.

9.0 SCHEDULED MAINTENANCE FOR SKYWATCH

Scheduled Maintenance	Daily	Weekly	Monthly	3x per Year	Annually
Hydraulic System[1]					
Weep Lines					
1. Make sure they are strapped to cylinders with wire ties				✓	
2. Make sure they are not cracked or damaged				✓	
3. Make sure they are connected to fittings at cylinder and pump				✓	
High Pressure Lines					
4. Inspect fittings at cylinders and pump for leaks				✓	
5. Inspect hose condition and check for leaks				✓	
Pump and Motor					
6. Check hydraulic fluid			✓		
7. Check pumps for leaks and loose connections				✓	
8. Check wiring on motor and verify proper operation				✓	
9. Inspect fill cap condition				✓	
10. Perform e-down function test			✓		
11. Check pump eliminator switch				✓	
Cylinders					
12. Clean piston shaft (in up position)				✓	
13. Check external seals for leaks (by raising and lowering tower twice)				✓	
14. Check for internal leaks (by removing weep lines at pump, then raising and lowering tower)				✓	

[1] Check for any leaks of hydraulic fluid and make notes of any maintenance concerns

Scheduled Maintenance, cont.	Daily	Weekly	Monthly	3x per Year	Annually
Generator					
Engine					
15. Check coolant	✓				
16. Check coolant anti-freeze protection every 500 hours[1]				✓	
17. Flush coolant / replace pressure cap every 1000 hours[1]					✓[2]
18. Replace spark plug (Gasoline models only) every 1000 hours[1]					✓[2]
19. Run test 15 minutes (check oil before starting generator)	✓				
20. Change engine oil every 150 hours[1] and replace oil filter [2, 7]				✓[3, 4, 5, 6]	
21. Replace air filter every 500 hours[1]				✓[4, 6]	
22. Replace fuel filter every 500 hours or as needed[1]					✓
23. Test breaker on generator				✓	
24. Check exhaust – is tail pipe secure?				✓	
25. Clean spark arrestor every 150 hours[1]				✓	

[1] The maintenance schedule for performance-based components (i.e. Scheduled engine maintenance) includes two numbers. One is based on the number of hours the component has been used, and the second is calendar-based. Maintenance intervals for these compartments should be whichever of the two numbers is lower.

[2] Perform at least once every two years

[3] As part of engine break-in, change the engine oil after the first 50 hours of operation on units that are not equipped with an OilMate system

[4] Perform more often when operating in dusty conditions

[5] Perform more often when operating in hot weather

[6] Perform at least once a year

[7] Oil change frequency is affected by OilMate system. See Appendix B for more information.

Scheduled Maintenance cont.	Daily	Weekly	Monthly	3x per Year	Annually
Generator					
Engine					
26. Replace engine V-belt every 1000 hrs[1] (Must be performed by a qualified mechanic authorized by Onan)					✓ 2,3
27. Clean crankcase breather every 1000 hrs[1] (Must be performed by a qualified mechanic authorized by Onan					✓ 2,3
28. Replace coolant hoses and thermostat every 1000 hrs[1] (Must be performed by a qualified mechanic authorized by Onan)					✓ 2,3
Other Components					
29. Test remote start from officer's compartment				✓	
30. Check generator wiring harness for damage				✓	
31. Check for fuel tank leaks			✓		
32. Verify fuel sending unit wires secured				✓	
33. Verify fuel lines secured				✓	
34. Examine Fuel Cap				✓	

[1] The maintenance schedule for performance-based components (i.e. Scheduled engine maintenance) includes two numbers. One is based on the number of hours the component has been used, and the second is calendar-based. Maintenance intervals for these compartments should be whichever of the two numbers is lower.

[2] Perform at least once every two years

[3] Must be performed by a qualified mechanic

Scheduled Maintenance cont.	Daily	Weekly	Monthly	3x per Year	Annually
Electrical System					
Solar Charging System					
35. Test generator start battery (check voltage)				✓	
36. Clean solar panel and check for damage			✓		
37. Test solar panel charger				✓	
38. Are wires running to panel and start battery secure?				✓	
12 Volt System					
Officer's Compartment					
39. Test interior light				✓	
40. Test fuel gauge				✓	
41. Test door relay switch / safety switch				✓	
42. Test 12 Volt outlet				✓	
43. Test flasher relay				✓	
44. Test descent alarm				✓	
45. Test strobe light				✓	
46. Verify Digital Windspeed Meter is functioning properly				✓	
Electrical Compartment					
47. Test interior light				✓	
48. Are battery cables secure and routed properly?				✓	
49. Check battery cables for damage				✓	
50. Test 12 Volt DC circuit breakers				✓	

Scheduled Maintenance cont.	Daily	Weekly	Monthly	3x per Year	Annually
110V AC System from Shore Power or Generator Power					
Officer's Compartment					
51. Test GFI Outlet				✓	
52. Perform air conditioner function test				✓	
53. Perform toe heater function test				✓	
54. Test camera				✓	
55. Test DVR				✓	
56. Test monitor				✓	
57. Test pan / tilt for camera				✓	
58. Test converter system (12 Volt output)				✓	
59. Test public address system				✓	
60. Test spotlights				✓	
61. Test floodlights				✓	
62. Test fuel gauge light				✓	
Electrical Compartment					
63. Test GFI outlet				✓	
64. Test Onan switch			✓		
65. Check wiring for damage				✓	

Scheduled Maintenance cont.	Daily	Weekly	Monthly	3x per Year	Annually
Pins, Pivots and Outriggers					
Grease Fittings					
Left Side					
66. Two points on Officer's Compartment at arm connection				✓	
67. One point at front cylinder				✓	
68. Two points at base cylinder				✓	
69. One point at upper arm pivot				✓	
70. One point at lower arm pivot				✓	
71. One point at left side front outrigger				✓	
72. One point at left side rear outrigger				✓	
Right Side					
73. Two points on Officer's Compartment at arm connection				✓	
74. One point at front cylinder				✓	
75. Two points at base of cylinder				✓	
76. One point at upper arm pivot				✓	
77. One point at lower arm pivot				✓	
78. One point at right side front outrigger				✓	
79. One point at right side rear outrigger				✓	
Pins					
80. Inspect top cylinder pins				✓	
81. Inspect bottom cylinder pins				✓	
82. Inspect upper arm pins				✓	
83. Inspect lower arm pins				✓	
84. Inspect Officer's Compartment pins				✓	
Outriggers					
85. Check outrigger pins to ensure they are secure and in good working order	✓				
86. Inspect outrigger plate bolts				✓	
87. Inspect outrigger jacks and bolts connecting them to the outrigger				✓	
88. Check outrigger reflective tape				✓	
Tongue					
89. Check split-ball coupler					
90. Check safety pins					
91. Check tongue jack					
92. Is breakaway brake switch, cable and 'S' hook intact?					

Scheduled Maintenance cont.	Daily	Weekly	Monthly	3x per Year	Annually
Hardware					
93. Are exterior door handles on Officer's Compartment operational?				✓	
94. Are interior door handles on Officer's Compartment operational?				✓	
95. Are latches in Officer's Compartment working smoothly?				✓	
96. Are paddle or twist handle door latches in good order?				✓	
97. Is generator service access door handle working correctly?				✓	
Paint					
98. Check for any defects in paint or primer				✓	
99. Are there any visible signs of rust?				✓	
100. Is anti-skid tape still in good shape?				✓	
101. Check under-coating on Officer's Compartment				✓	
Miscellaneous Inspections					
102. Are bumpers on bottom of Officer's Compartment in good working order?			✓		
103. Check air pressure in tires[1]			✓		
104. Inspect condition of spare tire					✓
105. Inspect wheel hubs				✓	
106. Is UHMW on outrigger plates damaged?				✓	
107. Check bubble levels				✓	
108. Are warning labels still visible?				✓	
109. Inspect weather stripping on Electrical Compartment				✓	
110. Inspect door safety cable				✓	
111. Inspect windows (cracks, tint, locks)				✓	
112. Inspect fire extinguisher				✓	
113. Is Officer's Compartment level?				✓	
114. If Applicable, check batteries and wind speed meter			✓		
115. Is User Guide up to date?					✓

[1]See stickers inside Electrical Compartment for tire inflation information

10.0 OBTAINING SUPPORT OR PARTS

When your SkyWatch Tower requires service, contact FLIR Integrated Systems Customer Support Center for assistance. Factory trained representatives are prepared to meet your support needs.

Please call Toll Free + 1.800.762.4796 or visit our website at http://www.flir.com/skywatch.

When contacting customer support, please provide the complete serial number listed on the data plate that is located on the tongue of the Tower.

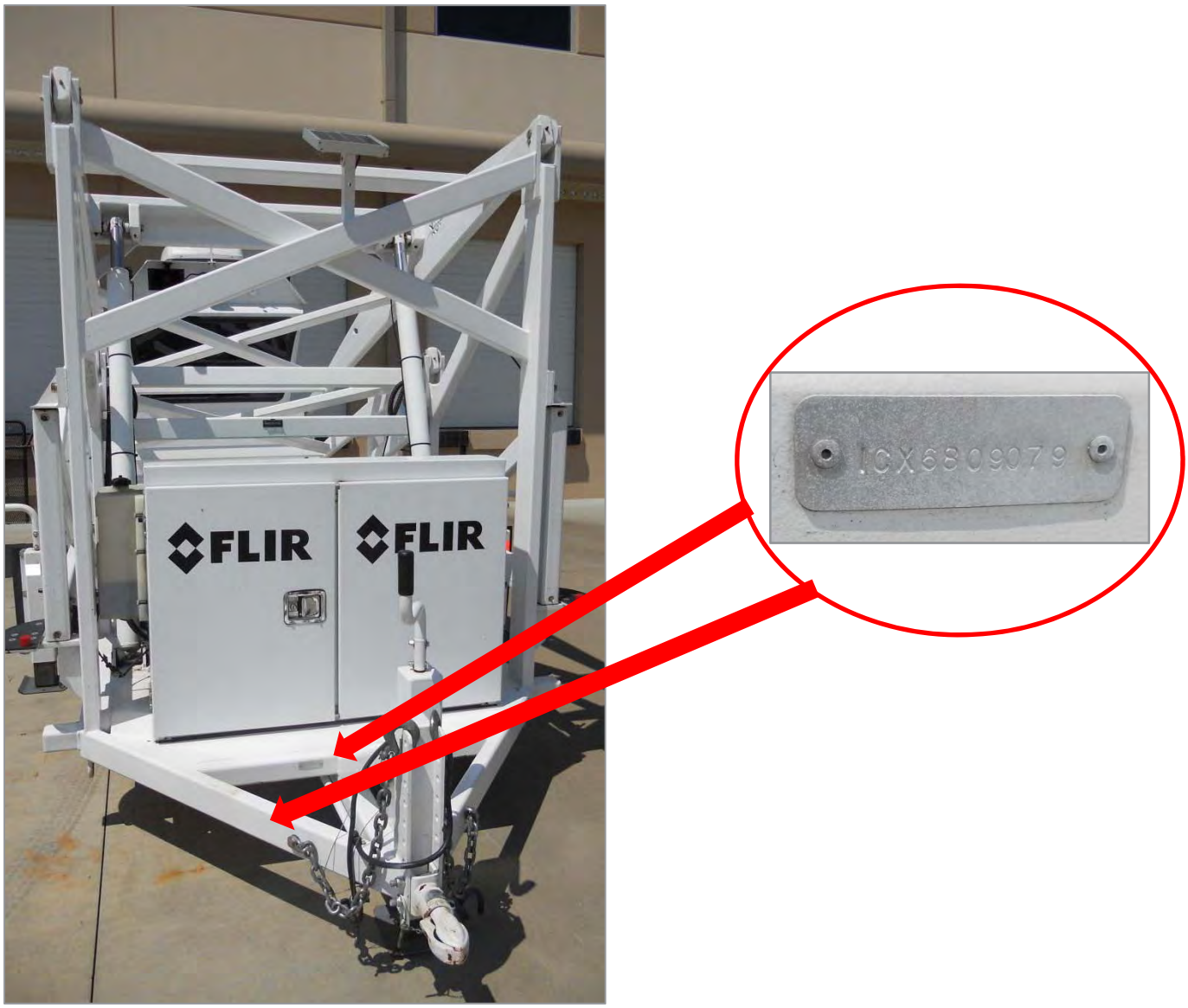


Figure 2 SkyWatch Serial Number locations

The serial number of the SkyWatch tower will be located in one of two locations, the arrows above indicate these locations.

APPENDIX A: OUTRIGGER LEVELING INSTRUCTIONS

Step 1: Swing all outriggers to their maximum extension and lock them in place.

Step 2: Release the drop leg extensions so that they are either fully extended or in contact with the ground. If they are not in contact with the ground at full extension, rotate the jack handle clockwise until contact with the ground is made.

Step 3: Give each jack handle a turn clockwise to ensure solid ground contact.

Step 4: Locate the three leveling bubbles on the tower.

A. The levels used to level to tower front-to-back are located at the base of each lower arm support column.

B. The level used to level the tower left-to-right is located on the cross beam directly above the Generator.

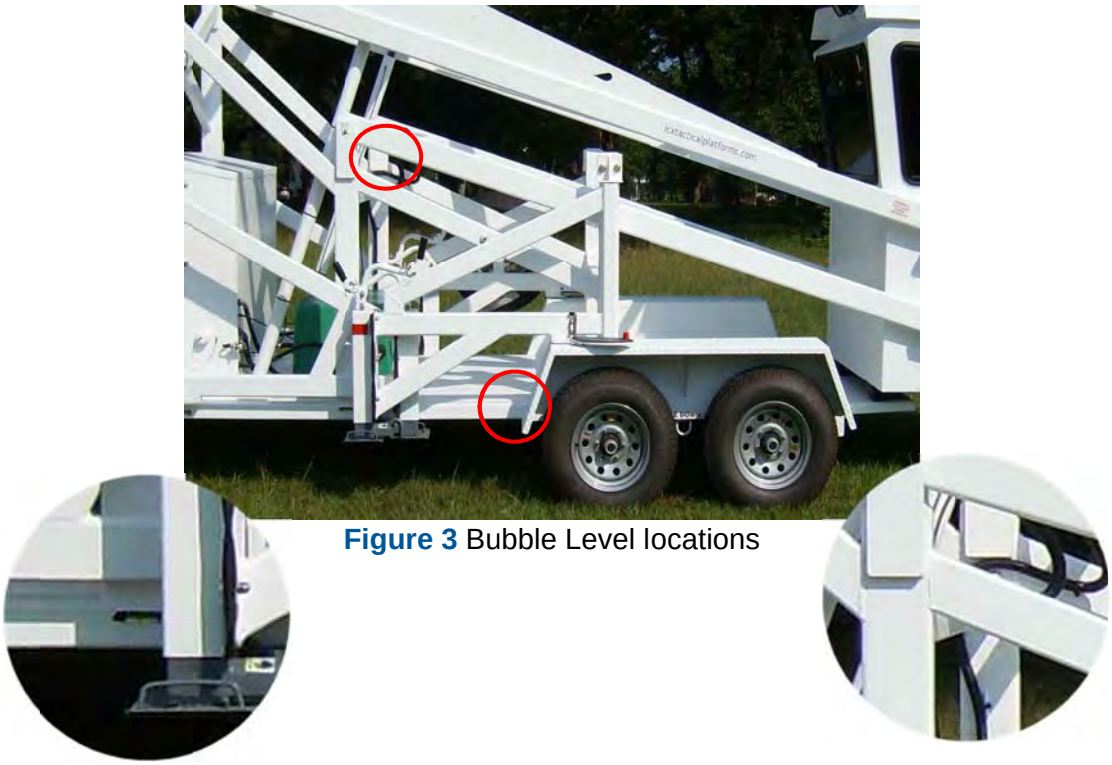


Figure 3 Bubble Level locations

Step 5: Using the tongue jack, slowly adjust the height of the front of the tower to bring the front-to-back leveling bubble into level.

Step 6: Next, use the outrigger jacks to bring the left-to-right leveling bubbles into level. Adjust the jacks in pairs, diagonally, for these adjustments.

APPENDIX B: OILMATE

Your SkyWatch Tower may come equipped with an OilMate system, installed on your generator. This system can substantially reduce the number of oil filter replacements and oil changes required to keep your generator running smoothly.

OilMate Usage Calculations for 5.5/5.0 KW Generator

OilMate Reservoir	5 gallons
OilMate oil usage	1 ounce/80 minutes run time



OilMate oil reservoir – Do NOT operate generator if reservoir is empty! The OilMate system will exhaust the generator's internal reserves and may damage the generator.

Water Separator – to keep the generator running at peak condition to more efficiently remove the waste oil, this component removes water which may be present in your diesel fuel. Check the fluid levels in the bowl periodically and open the drain to remove the water when it fills more than half of the bowl.

Oilmate Oil Usage Requirements Worksheet

Hours of Generator Run Time Per Week	Initial Reservoir Fill Level (quarts)	Oil Usage Annually (quarts)	Refill Requirements[1, 2]	Total Generator Runtime (Annually)	Filter Change / Oil Change (Annual Recommendations)
5[3]	10	6.09375	None (Refill Annually)	260	1 Oil Filter change Oil change not needed
10	12	12.1875	Add 1 qt. every month	520	1 Oil Filter change Oil change not needed
12	16	14.625	None (Refill Annually)	624	2 Oil Filter changes Oil change not needed
15	16	18.2813	Add 3 gts. every other month	780	2 Oil Filter changes Oil change not needed
20	18	24.375	Add 2 qts. every month	1040	2 Oil Filter changes Oil change not needed
40	18	48.75	Add 1 qt. every week	2080	4 Oil Filter changes Oil change not needed
80	18	97.5	Add 2 qts. every week	4160	8 Oil Filter changes 1 Oil change needed

NOTE: Oil changes are recommended after 4000 hours of runtime.

[1] When refilling reservoir, do not allow oil to overflow, environmental conditions may cause unit to use less oil than noted

[2] Check reservoir at least weekly and add oil as needed, environmental conditions may cause unit to use more oil than noted

[3] This assumes 5 hours of continuous use at one time. If GenSet is run 1 hour/day, OilMate system will not function and regular oil filter change and change intervals should be observed (changes recommended every 150 of runtime).

APPENDIX C: THE SKYWATCH PAINT PROCESS

The Tower structure and components are prepared for the paint process by a vigorous and thorough sand blasting. Next, these components are cleaned and carefully inspected for any defects before proceeding to the next step - caulk application. For this we use a styrenated siliconized acrylic caulk to seal adjoining sheet metal seams, preventing any moisture from entering these areas. In addition to helping to prevent rust creep, this application process also cosmetically enhances the appearance of these jointed areas. The entire structure is again inspected for defects and any found are corrected before proceeding to the primer coating. Here a 3.0 to 4.5 wet mil coating of an industrial and marine grade primer coating is applied. The primer coating we use is an inorganic reinforced zinc-rich primer formulated to cathodically protect steel in various industrial and marine environments. We then apply 4.5 to 9.0 wet mils of an acrylic polyurethane top coating. Our top coating is a polyester-modified aliphatic especially suited for industrial environments that provides a color and gloss fast retention for exterior exposure. Our coating conforms to ANSI/AWWA D102-03 OCS #5 & #6. After the final coating a rigorous inspection is performed to insure that a quality and lasting paint coating has been achieved.

Care and Maintenance of SkyWatch Paint Coatings

Wash units with solution of mild detergent and water every 2 weeks and then rinse thoroughly.

Wax units with a car wax product every 6 months.

Visual checks of the paint should be made on either a weekly or monthly basis to check for any nicks or rust spots. Any detected should be sanded and painted over with a primer and top coat.

Due to the cured film hardness of the SkyWatch top coating, we do not suggest that any clear coat products be applied after the Tower is delivered.

In the event of the appearance of rust due to a film break or atmospheric influence, contact the local Sherwin Williams store to arrange for inspection and preparation of specifications.

APPENDIX D: SKYWATCH WEIGHT LOAD RATINGS

icx™ tactical platforms

105 Forest Parkway
Air Logistics Center II
Suite 400
Forest Park, GA 30297
T + 1.770.850.1550
F + 1.770.850.1551
www.tacticalplatforms.icxt.com

January 01, 2005

SUBJECT: LOAD CAPACITY FOR SKYWATCH SINGLE MAN CAB TRAILERS

The rated cab loading capacity for all SkyWatch trailers employing single man cabs is 500 pounds based on ICx and third party vendor testing.

The 500 pound cab payload rating includes personnel and any equipment and/or material not included with the purchase of the trailer surveillance system.

END OF LETTER

Randall M. Bachtel, P.E.
Engineering Department Manager



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April 09, 2009

SUBJECT: LOAD CAPACITY FOR SKYWATCH TWO (2) MAN CAB TRAILERS

The rated cab loading capacity for all SkyWatch trailers employing two (2) man cabs is 800 pounds based on ICx testing and evaluation.

The 800 pound cab payload rating includes personnel and any equipment and/or material not included with the purchase of the trailer surveillance system.

END OF LETTER

Randall M. Bachtel, P.E.
Engineering Department Manager

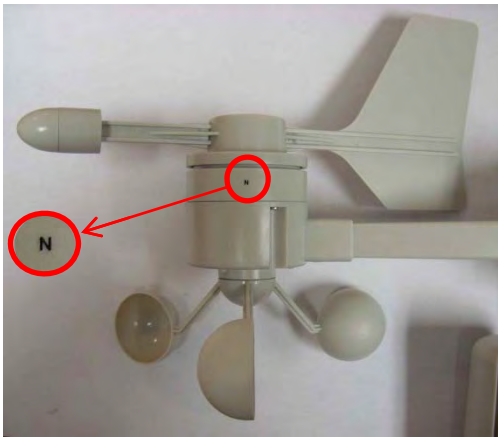


APPENDIX E: WEATHER STATION

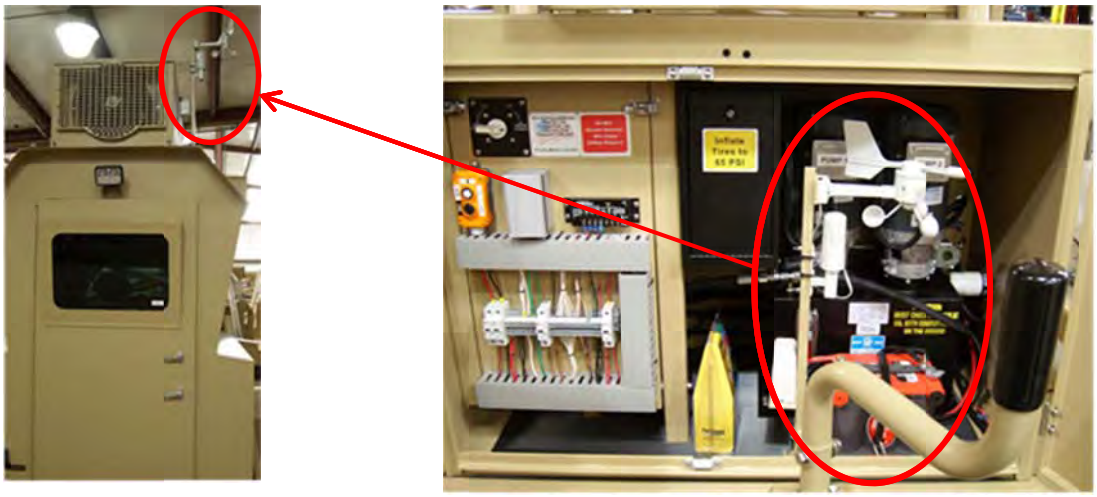


Your Tower may be equipped with a Weather Station. This wireless weather monitoring system is designed so that the Sensor Array Assembly is deployed along with the Tower and then stowed in the Electrical Compartment during Tower transport.

Deployment of the Sensor Array Assembly



Remove the Sensor Array Assembly from its storage location inside the Electrical Compartment and install it on the mounting post on top of the Officer's Compartment. Twist the Assembly so that the "N" on the anemometer faces north. Tighten the thumbscrews to lock the orientation of the Assembly. Remember to remove the Assembly from the Officer's Compartment before moving the Tower. Always store the Assembly in the Electrical Compartment during transport and when the Tower is not in use.



Battery Requirements

The Display Panel requires 3 "AA" batteries, the Sensor Array Assembly Transmitter requires 2 "AA" batteries and the Sensor Array Assembly Rain Gauge requires 2 "AAA" batteries.

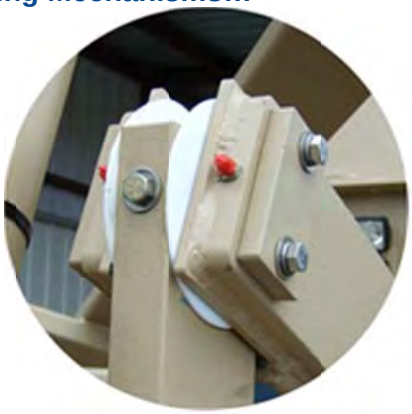
APPENDIX F: SPECIAL PRECAUTIONS

Our heaviest Towers, the Frontier Level IV and the Frontier Level III 2-Man models, have a special mechanism installed to help adjust the raising arms for such heavy payloads. This mechanism should **NEVER** be adjusted by the end user. A total of four (4) of these adjusting mechanisms are installed in each Tower.



Figure 4 2012 SkyWatch Frontier Level IV

Adjusting Mechanisms…



On each arm



As seen from above

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